Worksheet — Drag and Stokes' Law Simulation

Use with *Simulation - Drag and Stokes law.html* · goes with Station 5 (race to the bottom)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/07-drag-stokes** in a browser (offline: *Simulation - Drag and Stokes law.html*). **Ball in a liquid**: ▶️ Drop, and the stopwatch times the ball between bands A and B (20.0 cm apart). Use ×5 or ×20 for slow runs. **Skydiver**: ▶️ Jump, then ☂️.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab, Station 5 (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use g = 9.81 N kg^{-1}. Stokes' law: $F_d = 6\pi\eta rv$. Glycerol: ρ = 1260 kg m^{-3}, η = 1.41 Pa s. Steel: ρ = 7850 kg m^{-3}.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. What happens to the speed of the ball after it is let go?			
2. At terminal velocity, what is the resultant force?			
3. You double the radius of the ball. Terminal speed?			
4. The same steel ball in water instead of glycerol?			

5. Draw the free-body diagram of the ball at terminal velocity. Then show that $v_t = 2r^2(\rho_{ball} - \rho_{fluid})g / (9\eta)$.

Free-body diagram and derivation

6. Calculate v_t for the steel ball (r = 1.0 mm) in glycerol.

Working

Part 2 — During the lab: Station 5

ball	*r* / mm	sim t_{AB} / s	our *t* / s (3 trials)	our mean *t* / s	$v = 0.200/t$ / cm s^{-1}
small	1.0				
large	2.0				

7. By what factor is the large ball faster? What does Stokes' law predict? Why is the sim's ratio a little less than 4?

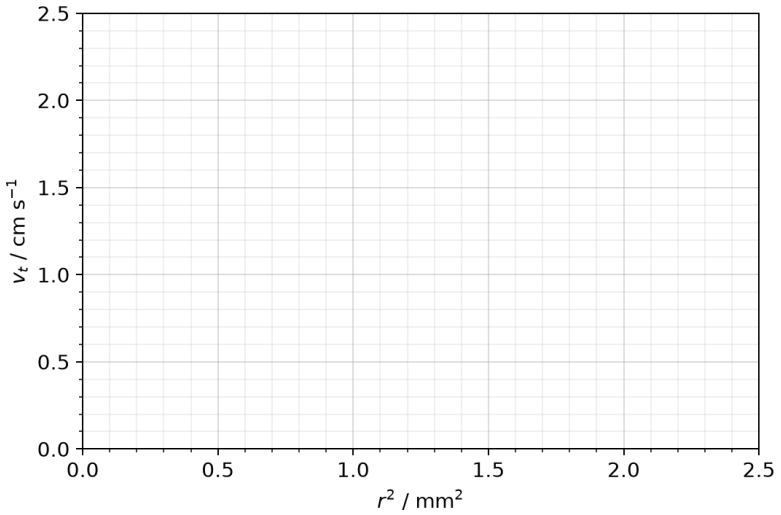
8. How could you check that the ball has reached terminal speed before band A?

Part 3 — Homework: viscosity from a graph

r / mm	0.50	0.75	1.00	1.25	1.50
r^2 / mm^2					
t_{AB} / s					

v_t / cm s^{-1}					

9. Plot v_t against r^2 and draw the best-fit line through the origin.



10. Use the gradient to find the viscosity of glycerol (convert the gradient to m^{-1} s^{-1} first).

Working

11. Sketch the *v*–*t* graph of a skydiver who opens a parachute after 18 s. Explain each part using the forces. Why can't Stokes' law be used for the skydiver?

Sketch and explanation